

SOC-5811 Week 7: Regression with multiple variables

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1/45



REVIEW CATEGORICAL PREDICTOR VARIABLES

```
penguin_df <-  
  palmerpenguins::penguins %>%  
    na.omit()  
head(penguin_df)
```

```
## # A tibble: 6 x 8  
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g  
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>  
## 1 Adelie  Torgersen         39.1          18.7          181          3750  
## 2 Adelie  Torgersen         39.5          17.4          186          3800  
## 3 Adelie  Torgersen         40.3          18           195          3250  
## 4 Adelie  Torgersen         36.7          19.3          193          3450  
## 5 Adelie  Torgersen         39.3          20.6          190          3650  
## 6 Adelie  Torgersen         38.9          17.8          181          3625  
## # i 2 more variables: sex <fct>, year <int>
```

REVIEW CATEGORICAL PREDICTOR VARIABLES

What happens if we use a categorical predictor variable with **two** levels?

$$y_i = \beta_0 + \beta_1(\text{Sex})$$



REVIEW CATEGORICAL PREDICTOR VARIABLES

```
summary(lm(bill_depth_mm ~ sex, data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ sex, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.7911 -1.8911  0.5745  1.3745  4.2745
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   16.4255     0.1425 115.286 < 2e-16 ***
## sexmale       1.4656     0.2006   7.307 2.07e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.83 on 331 degrees of freedom
## Multiple R-squared:  0.1389, Adjusted R-squared:  0.1363
## F-statistic: 53.39 on 1 and 331 DF,  p-value: 2.066e-12
```

REVIEW CATEGORICAL PREDICTOR VARIABLES

```
penguin_df <- penguin_df %>%  
  dummy_cols('sex')  
penguin_df %>%  
  select(species, bill_length_mm, bill_depth_mm, sex, sex_female, sex_male)
```

```
## # A tibble: 333 x 6  
##   species bill_length_mm bill_depth_mm sex    sex_female sex_male  
##   <fct>      <dbl>         <dbl> <fct>    <int>     <int>  
## 1 Adelie      39.1           18.7 male      0         1  
## 2 Adelie      39.5           17.4 female    1         0  
## 3 Adelie      40.3           18   female    1         0  
## 4 Adelie      36.7           19.3 female    1         0  
## 5 Adelie      39.3           20.6 male      0         1  
## 6 Adelie      38.9           17.8 female    1         0  
## 7 Adelie      39.2           19.6 male      0         1  
## 8 Adelie      41.1           17.6 female    1         0  
## 9 Adelie      38.6           21.2 male      0         1  
## 10 Adelie     34.6           21.1 male      0         1  
## # i 323 more rows
```

REVIEW CATEGORICAL PREDICTOR VARIABLES

```
summary(lm(bill_depth_mm ~ sex_male, data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ sex_male, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.7911 -1.8911  0.5745  1.3745  4.2745
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.4255      0.1425 115.286 < 2e-16 ***
## sex_male      1.4656      0.2006   7.307 2.07e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.83 on 331 degrees of freedom
## Multiple R-squared:  0.1389, Adjusted R-squared:  0.1363
## F-statistic: 53.39 on 1 and 331 DF,  p-value: 2.066e-12
```

REVIEW CATEGORICAL PREDICTOR VARIABLES

- Do I need to add both dummy variables to my regression model? Why or why not?

```
penguin_df %>%  
  select(species, bill_length_mm, bill_depth_mm, sex, sex_female, sex_male)
```

```
## # A tibble: 333 x 6  
##   species bill_length_mm bill_depth_mm sex    sex_female sex_male  
##   <fct>      <dbl>         <dbl> <fct>      <int>      <int>  
## 1 Adelie      39.1           18.7 male         0          1  
## 2 Adelie      39.5           17.4 female        1          0  
## 3 Adelie      40.3           18   female        1          0  
## 4 Adelie      36.7           19.3 female        1          0  
## 5 Adelie      39.3           20.6 male          0          1  
## 6 Adelie      38.9           17.8 female        1          0  
## 7 Adelie      39.2           19.6 male          0          1  
## 8 Adelie      41.1           17.6 female        1          0  
## 9 Adelie      38.6           21.2 male          0          1  
## 10 Adelie     34.6           21.1 male          0          1  
## # i 323 more rows
```

REVIEW CATEGORICAL PREDICTOR VARIABLES

```
summary(lm(bill_depth_mm ~ sex_male + sex_female, data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ sex_male + sex_female, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.7911 -1.8911  0.5745  1.3745  4.2745
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.4255      0.1425 115.286 < 2e-16 ***
## sex_male      1.4656      0.2006   7.307 2.07e-12 ***
## sex_female      NA           NA      NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.83 on 331 degrees of freedom
## Multiple R-squared:  0.1389, Adjusted R-squared:  0.1363
## F-statistic: 53.39 on 1 and 331 DF, p-value: 2.066e-12
```


REVIEW MULTICOLLINEARITY

- ▶ The problem of **multicollinearity**.

REVIEW MULTICOLLINEARITY

- ▶ The problem of **multicollinearity**.
- ▶ In a regression model, you can't have predictors that are a perfect function of one another. In other words, you can't have perfect **multicollinearity** where predictors are perfectly correlated with one another.



REVIEW MULTICOLLINEARITY

- ▶ The problem of **multicollinearity**.
- ▶ In a regression model, you can't have predictors that are a perfect function of one another. In other words, you can't have perfect **multicollinearity** where predictors are perfectly correlated with one another.
- ▶ Why?



REVIEW

1. Choose a functional form for the model.

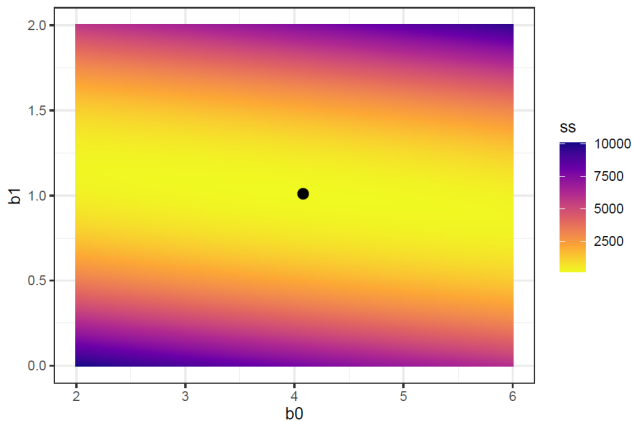
$$\text{depth}_i = \beta_0 + \beta_1 \text{sex_male} + \beta_2 \text{sex_female} + \epsilon_i$$

2. Choose an estimator (i.e., objective function) that relates the model to real observed data (e.g., OLS).

$$\text{Sum of squared residuals} = \sum_{i=1}^n (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$$

3. Fit the model (e.g., estimate parameters) by optimizing the objective function.

REVIEW MULTICOLLINEARITY



REVIEW MULTICOLLINEARITY

- ▶ If you provide a categorical predictor variable to `lm()`, the default behavior is to transform it into $N-1$ dummy variables where N is the number of unique levels.



REVIEW CATEGORICAL PREDICTOR VARIABLES

```
penguin_df <- penguin_df %>%  
  dummy_cols('species')  
head(penguin_df)
```

```
## # A tibble: 6 x 13  
##   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g  
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>  
## 1 Adelie  Torgersen         39.1          18.7          181          3750  
## 2 Adelie  Torgersen         39.5          17.4          186          3800  
## 3 Adelie  Torgersen         40.3          18           195          3250  
## 4 Adelie  Torgersen         36.7          19.3          193          3450  
## 5 Adelie  Torgersen         39.3          20.6          190          3650  
## 6 Adelie  Torgersen         38.9          17.8          181          3625  
## # i 7 more variables: sex <fct>, year <int>, sex_female <int>, sex_male <int>,  
## #   species_Adelie <int>, species_Chinstrap <int>, species_Gentoo <int>
```

REVIEW CATEGORICAL PREDICTOR VARIABLES

```
summary(lm(bill_depth_mm ~ species_Adelie + species_Gentoo + species_Chinstrap,
           data=penguin_df))
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ species_Adelie + species_Gentoo +
##     species_Chinstrap, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.84726 -0.79664  0.00336  0.75274  3.15274
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    18.42059     0.13627  135.181   <2e-16 ***
## species_Adelie  -0.07333     0.16497   -0.444    0.657
## species_Gentoo  -3.42395     0.17082  -20.044   <2e-16 ***
## species_Chinstrap      NA           NA      NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.124 on 330 degrees of freedom
## Multiple R-squared:  0.6764, Adjusted R-squared:  0.6744
## F-statistic: 344.8 on 2 and 330 DF,  p-value: < 2.2e-16
```


REVIEW CATEGORICAL PREDICTOR VARIABLES

```
penguin_df <- penguin_df %>%  
  mutate(species = factor(species, levels=c('Chinstrap', 'Adelie', 'Gentoo')))  
summary(lm(bill_depth_mm ~ species, data = penguin_df))
```

```
##  
## Call:  
## lm(formula = bill_depth_mm ~ species, data = penguin_df)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.84726 -0.79664  0.00336  0.75274  3.15274   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   18.42059    0.13627  135.181  <2e-16 ***  
## speciesAdelie -0.07333    0.16497   -0.444    0.657      
## speciesGentoo -3.42395    0.17082  -20.044  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 1.124 on 330 degrees of freedom  
## Multiple R-squared:  0.6764, Adjusted R-squared:  0.6744   
## F-statistic: 344.8 on 2 and 330 DF,  p-value: < 2.2e-16
```

REVIEW CATEGORICAL PREDICTOR VARIABLES

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$



REVIEW CATEGORICAL PREDICTOR VARIABLES

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$

$$y_i = (18.42) + (-0.07)(\text{Adelie}) + (-3.42)(\text{Gentoo})$$



REVIEW CATEGORICAL PREDICTOR VARIABLES

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$

$$y_i = (18.42) + (-0.07)(\text{Adelie}) + (-3.42)(\text{Gentoo})$$

$$E[Y|\text{species} = \text{Adelie}] = (18.42) + (-0.07)(1) + (-3.42)(0) = \mathbf{18.35}$$

REVIEW CATEGORICAL PREDICTOR VARIABLES

$$y_i = \beta_0 + \beta_1(\text{Adelie}) + \beta_2(\text{Gentoo})$$

$$y_i = (18.42) + (-0.07)(\text{Adelie}) + (-3.42)(\text{Gentoo})$$

$$E[Y|\text{species} = \text{Adelie}] = (18.42) + (-0.07)(1) + (-3.42)(0) = \mathbf{18.35}$$

$$E[Y|\text{species} = \text{Chinstrap}] = (18.42) + (-0.07)(0) + (-3.42)(0) = \mathbf{18.42}$$

MULTIPLE PREDICTORS

- ▶ The coefficient β_k is the average or expected difference in outcome y , comparing two people who differ by one unit in the predictor x_k while being equal in all the other predictors. This is sometimes stated in shorthand as comparing two people (or, more generally, two observational units) that differ in x_k with all the other predictors held constant.

MULTIPLE PREDICTORS

```
mod <- lm(bill_depth_mm ~ bill_length_mm + species, data = penguin_df)
summary(mod)
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm + species, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.4579 -0.6814 -0.0431  0.5441  3.5994
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.63218    0.87109   9.910 < 2e-16 ***
## bill_length_mm  0.20044    0.01768  11.337 < 2e-16 ***
## speciesAdelie   1.93308    0.22572   8.564 4.26e-16 ***
## speciesGentoo  -3.17024    0.14679 -21.598 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9543 on 329 degrees of freedom
## Multiple R-squared:  0.7673, Adjusted R-squared:  0.7652
## F-statistic: 361.6 on 3 and 329 DF,  p-value: < 2.2e-16
```

MULTIPLE PREDICTORS

- ▶ $\beta_0 = 8.63$: If a penguin is a Chinstrap penguin with a bill length of 0, then we would predict this penguin to have a bill depth of **8.63**.



MULTIPLE PREDICTORS

- ▶ $\beta_0 = 8.63$: If a penguin is a Chinstrap penguin with a bill length of 0, then we would predict this penguin to have a bill depth of **8.63**.
- ▶ $\beta_1 = 0.20$: Comparing penguins within the same species but whose bill lengths differ by 1, we would predict their difference in bill depths to be **0.20**.



MULTIPLE PREDICTORS

- ▶ $\beta_0 = 8.63$: If a penguin is a Chinstrap penguin with a bill length of 0, then we would predict this penguin to have a bill depth of **8.63**.
- ▶ $\beta_1 = 0.20$: Comparing penguins within the same species but whose bill lengths differ by 1, we would predict their difference in bill depths to be **0.20**.
- ▶ $\beta_2 = 1.93$: Comparing penguins who have the same bill length but one is Chinstrap and one is Adelie, we would predict their difference in bill depths to be **1.93**.



MULTIPLE PREDICTORS

- ▶ $\beta_0 = 8.63$: If a penguin is a Chinstrap penguin with a bill length of 0, then we would predict this penguin to have a bill depth of **8.63**.
- ▶ $\beta_1 = 0.20$: Comparing penguins within the same species but whose bill lengths differ by 1, we would predict their difference in bill depths to be **0.20**.
- ▶ $\beta_2 = 1.93$: Comparing penguins who have the same bill length but one is Chinstrap and one is Adelie, we would predict their difference in bill depths to be **1.93**.
- ▶ $\beta_3 = -3.17$: Comparing penguins who have the same bill length but one is Chinstrap and one is Gentoo, we would predict their difference in bill depths to be **3.17**.

REVIEW

- ▶ On the board: health as a function of income and wealth.



REVIEW

- ▶ On the board: health as a function of income and wealth.
- ▶ The coefficients of each variable are capturing variation *independent* of the other variable, or *marginal* to the other variable. To estimate the coefficient of income on health, we are using variation in our data that is mechanically unrelated to wealth; in other words, differences **cannot** be attributable to wealth because we are only comparing units with the same level of wealth.



REVIEW

► $\text{health} = \beta_0 + \beta_1 \text{income} + \beta_2 \text{wealth}$



REVIEW

- ▶ $\text{health} = \beta_0 + \beta_1 \text{income} + \beta_2 \text{wealth}$
- ▶ The effects of each variable are also *independent* of each other. My estimated coefficient of income is the same no matter what level of wealth I'm using. I'm averaging over the differences within any given level of wealth to arrive at an overall *average* coefficient for income on health.



REVIEW

- ▶ $\text{health} = \beta_0 + \beta_1 \text{income} + \beta_2 \text{wealth}$
- ▶ The effects of each variable are also *independent* of each other. My estimated coefficient of income is the same no matter what level of wealth I'm using. I'm averaging over the differences within any given level of wealth to arrive at an overall *average* coefficient for income on health.
- ▶ We may be averaging over interesting variation. We could imagine the effect of income on health *depends* on level of wealth. We may want to estimate an *interaction* between income and wealth.

MULTIPLE PREDICTORS

- ▶ Regression is a way of making comparisons; it is very important to understand and be able to articulate the comparison you are making.

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

MULTIPLE PREDICTORS

- ▶ Regression is a way of making comparisons; it is very important to understand and be able to articulate the comparison you are making.

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

$$\beta_1 = E[d|l = x + 1, s] - E[d|l = x, s] = \mathbf{0.20}$$

MULTIPLE PREDICTORS

- Regression is a way of making comparisons; it is very important to understand and be able to articulate the comparison you are making.

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

$$\beta_1 = E[d|l = x + 1, s] - E[d|l = x, s] = \mathbf{0.20}$$

$$\beta_2 = E[d|s = \text{Adelie}, l] - E[d|s = \text{Chinstrap}, l] = \mathbf{1.93}$$

MULTIPLE PREDICTORS

- ▶ Regression is a way of making comparisons; it is very important to understand and be able to articulate the comparison you are making.

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

$$\beta_1 = E[d|l = x + 1, s] - E[d|l = x, s] = \mathbf{0.20}$$

$$\beta_2 = E[d|s = \text{Adelie}, l] - E[d|s = \text{Chinstrap}, l] = \mathbf{1.93}$$

$$\beta_3 = E[d|s = \text{Gentoo}, l] - E[d|s = \text{Chinstrap}, l] = \mathbf{-3.17}$$

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

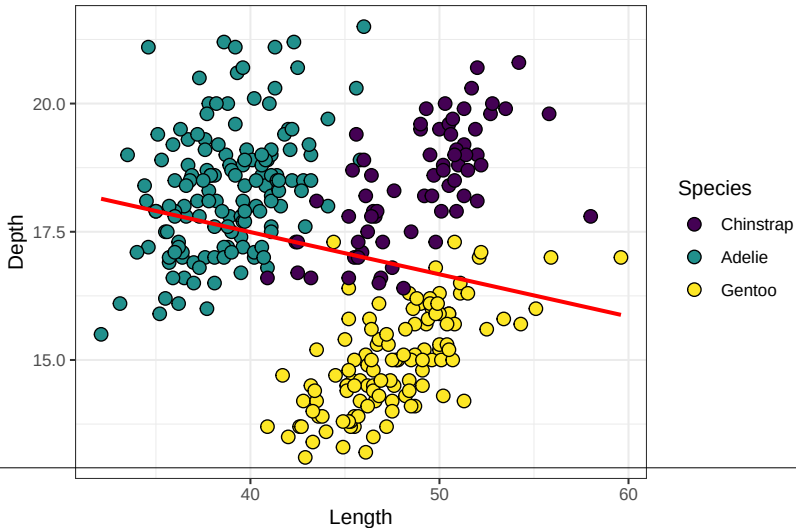
- How many estimates do we have of the association between bill length and depth?

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

- ▶ How many estimates do we have of the association between bill length and depth?
- ▶ We just have one estimate (β_1) regardless of whether we control for species.

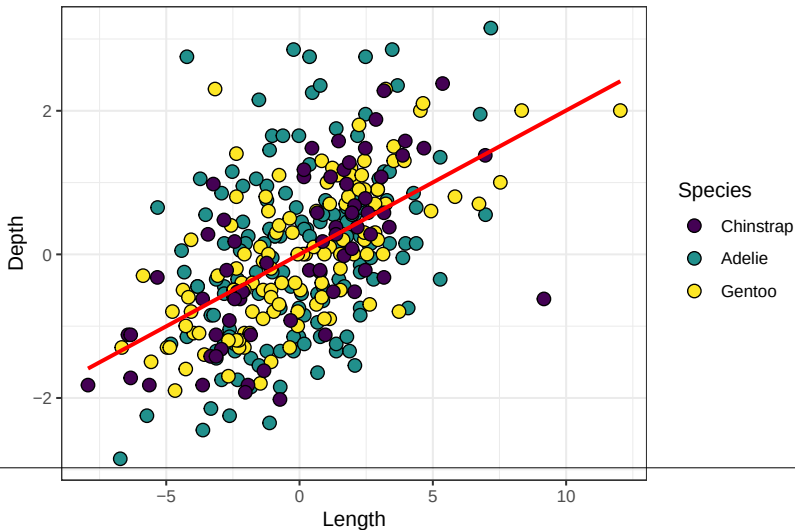
HOLDING SOMETHING CONSTANT

$$d_i = \beta_0 + \beta_1(l_i)$$



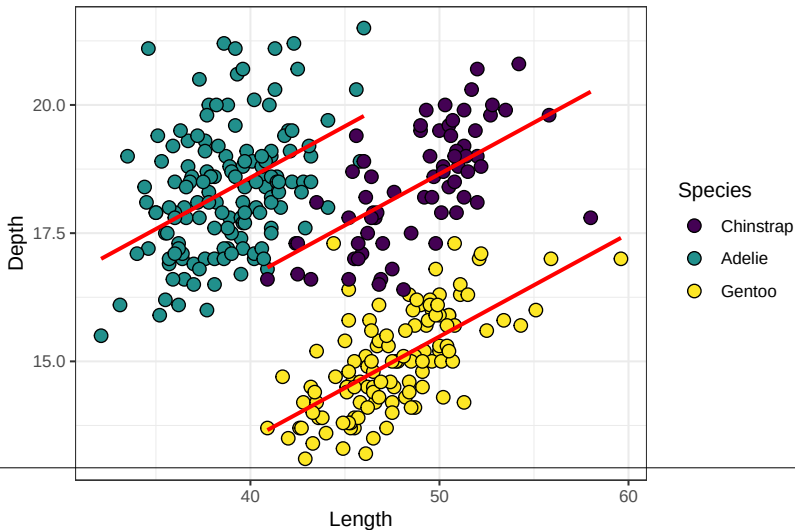
HOLDING SOMETHING CONSTANT

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$



MULTIPLE PREDICTORS

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$



INTERACTIONS

- In our model, the slope of the regression for bill length was forced to be equal across all subgroups defined by species. In other words, we just have one β_1 .

$$d_i = l_i + s_i$$

$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}})$$

INTERACTIONS

- Now how many coefficients do we have for length?

$$d_i = l_i + s_i + (l_i \times s_i)$$

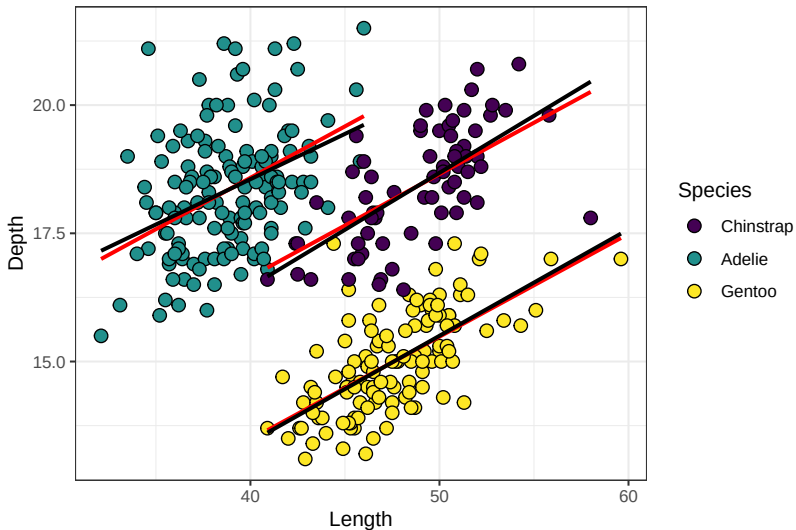
$$d_i = \beta_0 + \beta_1(l_i) + \beta_2(s_i^{\text{Adelie}}) + \beta_3(s_i^{\text{Gentoo}}) + \beta_4(l_i \times s_i^{\text{Adelie}}) + \beta_5(l_i \times s_i^{\text{Gentoo}})$$

INTERACTIONS

```
mod2 <- lm(bill_depth_mm ~ bill_length_mm * species, data = penguin_df)
summary(mod2)
```

```
##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm * species, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.6574 -0.6559 -0.0483  0.5203  3.4990
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.56914    1.71127   4.423 1.33e-05 ***
## bill_length_mm  0.22221    0.03496   6.356 6.96e-10 ***
## speciesAdelie   3.91857    2.06731   1.895  0.0589 .
## speciesGentoo  -2.44818    2.17972  -1.123  0.2622
## bill_length_mm:speciesAdelie -0.04553    0.04594  -0.991  0.3224
## bill_length_mm:speciesGentoo -0.01460    0.04499  -0.324  0.7458
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9556 on 327 degrees of freedom
## Multiple R-squared:  0.7681, Adjusted R-squared:  0.7645
## F-statistic: 216.6 on 5 and 327 DF,  p-value: < 2.2e-16
```

INTERACTIONS



36/45



INTERACTIONS

```
mod3 <- lm(bill_depth_mm ~ bill_length_mm * body_mass_g, data = penguin_df)
summary(mod3)

##
## Call:
## lm(formula = bill_depth_mm ~ bill_length_mm * body_mass_g, data = penguin_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.8374 -1.1902 -0.0181  1.2037  4.1935
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.367e+01  4.411e+00   3.099  0.00211 **
## bill_length_mm  1.920e-01  9.640e-02   1.992  0.04720 *
## body_mass_g    7.061e-04  1.134e-03   0.623  0.53384
## bill_length_mm:body_mass_g -4.222e-05  2.409e-05  -1.752  0.08067 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.732 on 329 degrees of freedom
## Multiple R-squared:  0.2337, Adjusted R-squared:  0.2267
## F-statistic: 33.45 on 3 and 329 DF,  p-value: < 2.2e-16
```

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- ▶ “Holding constant” = “Comparing two items i and j that differ by an amount x on predictor k but are identical on all other predictors, the predicted difference $y_i - y_j$ is $B_k(x)$, on average.”



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Different research questions:

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- ▶ What is the difference in the average association between grandparent wealth and grandchild wealth by race?

CONFOUNDING

► Review Simpson's Paradox

Department	All		Men		Women	
	Applicants	Admitted	Applicants	Admitted	Applicants	Admitted
A	933	64%	825	62%	108	82%
B	585	63%	560	63%	25	68%
C	918	35%	325	37%	593	34%
D	792	34%	417	33%	375	35%
E	584	25%	191	28%	393	24%
F	714	6%	373	6%	341	7%
Total	4526	39%	2691	45%	1835	30%

Legend:

 greater percentage of successful applicants than the other gender

 greater number of applicants than the other gender

bold - the two 'most applied for' departments for each gender

CONFOUNDING

$$\text{Admission} = \beta_0 + \beta_1(\text{Gender})$$

$$\text{Admission} = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{Department})$$



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- ▶ There may be very good reasons to control for X! But it *completely* depends on your research question; more specifically, exactly what **comparison** you say you’re trying to make.
- ▶ We are transitioning to causal inference and the concept of *confounding*.



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- ▶ People who drink more wine are less likely to die ($p < 0.05$)!
- ▶ People who drank more wine are less likely to die *because* they drank more wine; *if* people drink more wine, *then* they will be less likely to die.

CONFOUNDING

$$\text{mortality rate} = \beta_0 + \beta_1 \text{wine} + \beta_2 \text{age} + \beta_3 \text{income} + \beta_4 \text{education} + \beta_5 \text{social}$$

- ▶ People who drink more wine are less likely to die comparing within levels of age, income, education, and social networks ($p < 0.05$)!

CONFOUNDING

$$\text{mortality rate} = \beta_0 + \beta_1 \text{wine}$$

How might we avoid having to control for so many things? In other words, how might we guarantee the *only* difference across our comparisons of drinking is the drinking?

